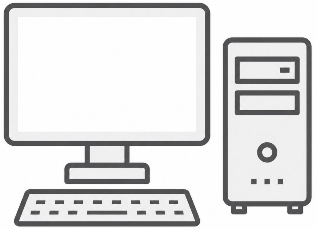
Prototyping Assignment: AI Image Classification System

Gökay Sengün | Lorenz Pusch | 3S | 13. Juli 2026

Application

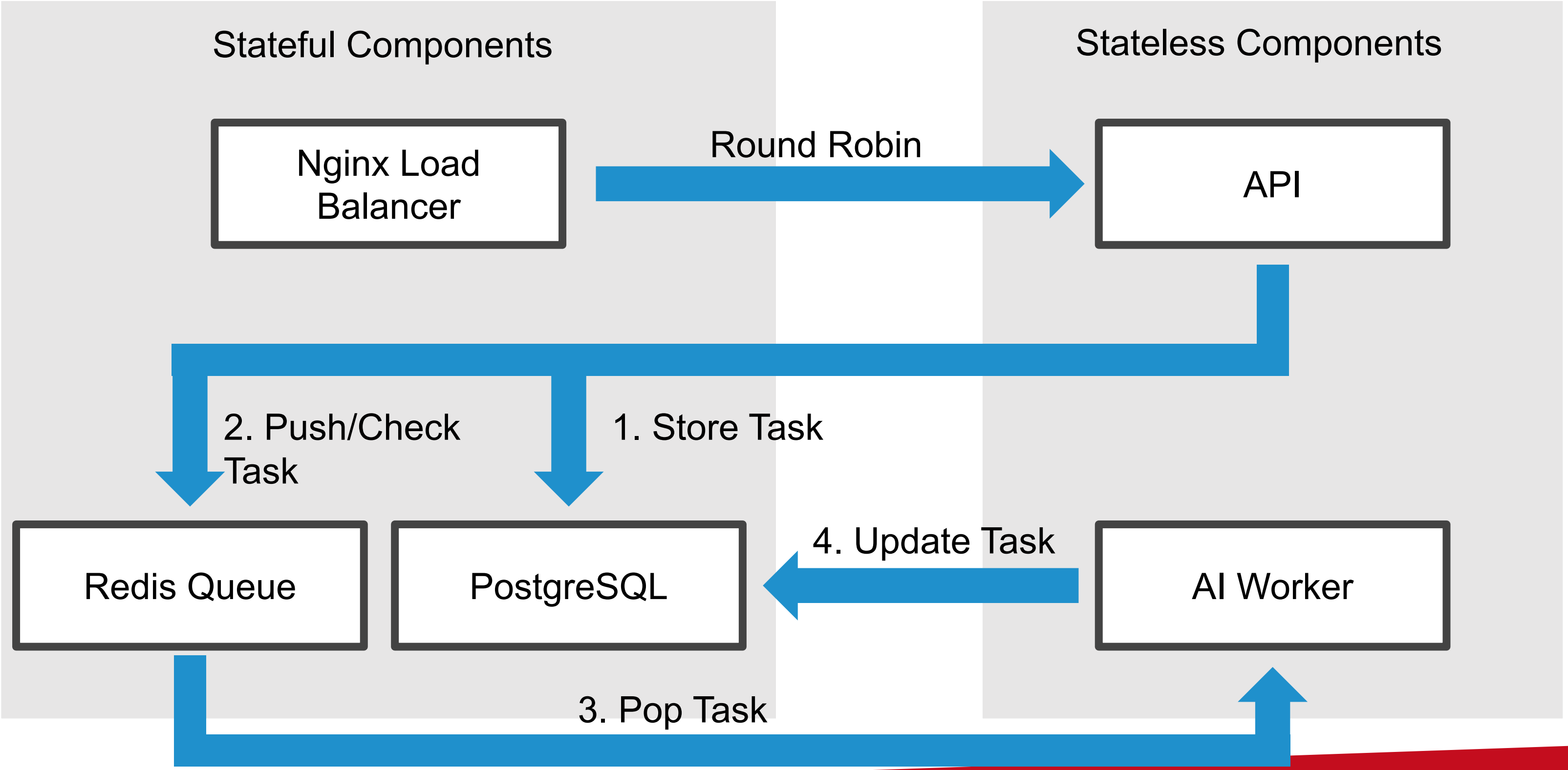


Golden Retriever

```
gokaysengun@MacBook-Air-von-Gokay ~ % curl -X POST http://34.185.188.250/classify \
        -H "Content-Type: application/json" \
        -d "{\"image_url\": \"https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS2FUZMIkoOnHgembMF9InnnlEXenXekksJ
rA&s\", \"task_id\": \"$MY_TASK_ID\"}"
{"task_id": "A6683AD3-08CA-4DC6-8CFB-9579FE7E82F2", "message": "Task queued"}%
```

```
gokaysengun@MacBook-Air-von-Gokay ~ % curl http://34.185.188.250/status/$MY_TASK_ID
{"task_id":"A6683AD3-08CA-4DC6-8CFB-9579FE7E82F2","status":"completed","result":"golden_retriever","created_at":"2026-
07-05T13:33:55.979112+00:00","finished_at":"2026-07-05T13:33:56.131783+00:00"}%
```

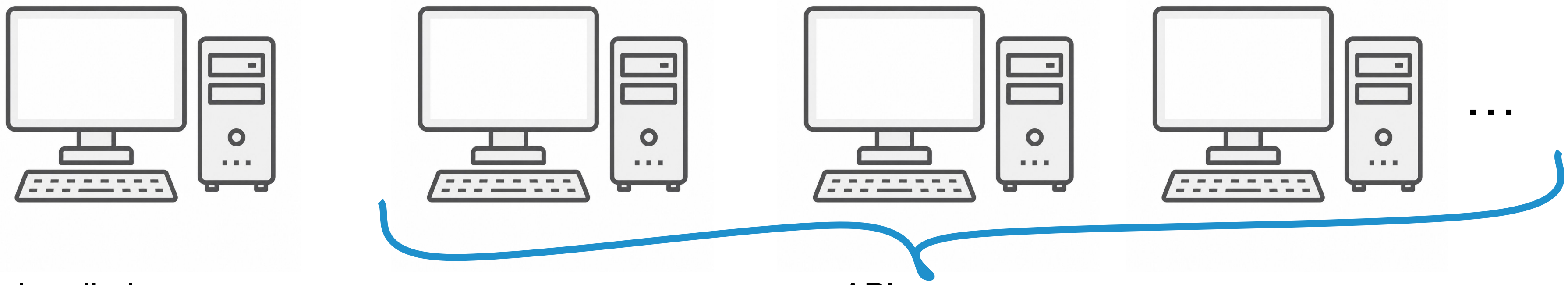
Architecture



Horizontal Scaling

Node 1

Node 2, Node 3, …



Loadbalancer
Redis Queue
PostgreSQL

API
PostgreSQL
AI Worker

Load Shedding & Multi-Tenant Fairness

```python
# from app/api/main.py
CLIENT_SOFT_LIMIT          = GLOBAL_MAX_REQUESTS * 0.2
CLIENT_BURST_LIMIT         = GLOBAL_MAX_REQUESTS * 0.5
GLOBAL_THROTTLE_THRESHOLD = GLOBAL_MAX_REQUESTS * 0.8

if client_count >= CLIENT_BURST_LIMIT:
    return Response(content="Client burst quota exceeded.", status_code=429)

if client_count >= CLIENT_SOFT_LIMIT and active_requests >= GLOBAL_THROTTLE_THRESHOLD:
    return Response(content="System under heavy load. Client soft quota exceeded.", status_code=429)
```

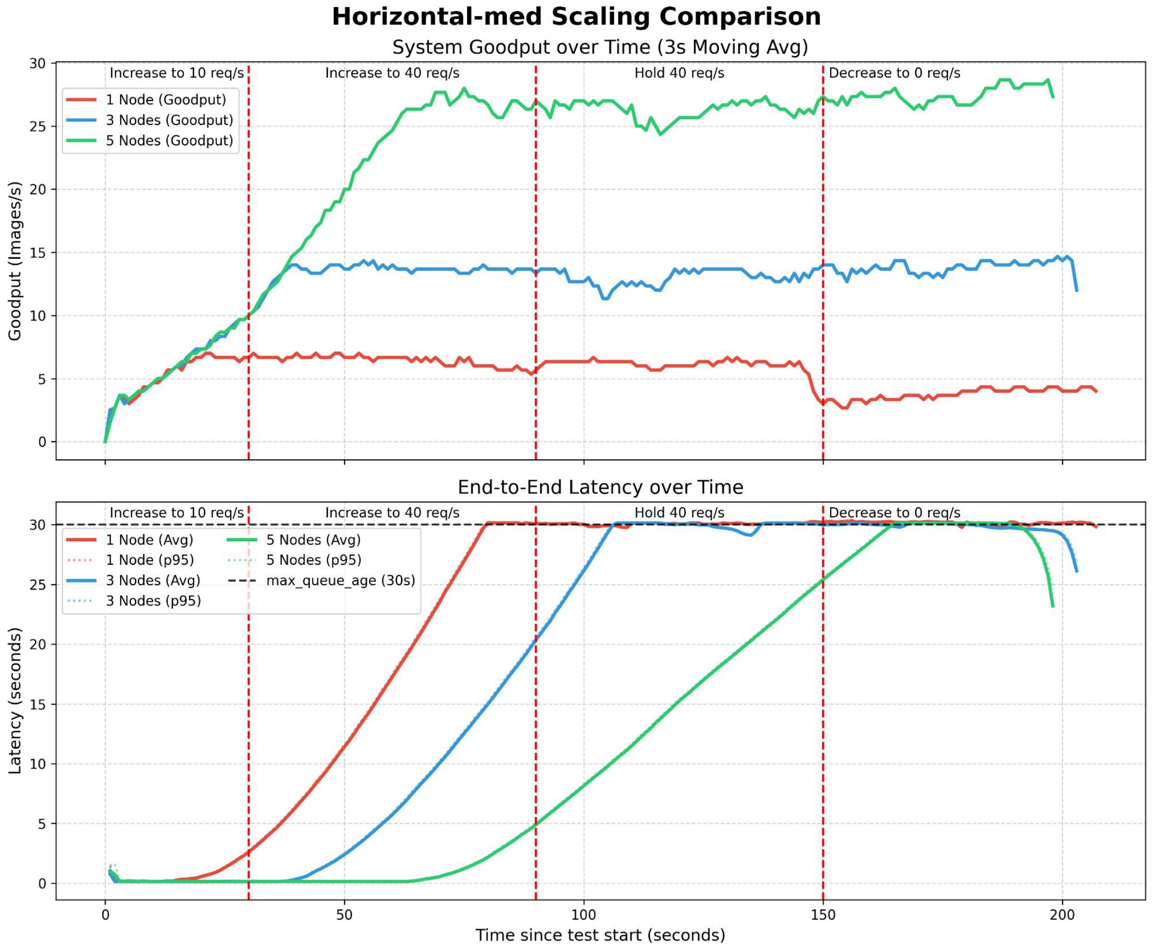
Backpressure

According to Little's Law, an insurmountable queue backlog leads to wasted work

```
# from infra/startup.sh.tpl
MAX_GLOBAL_QUEUE_SIZE=$(( THROUGHPUT * MAX_QUEUE_AGE * NUM_WORKER))
```

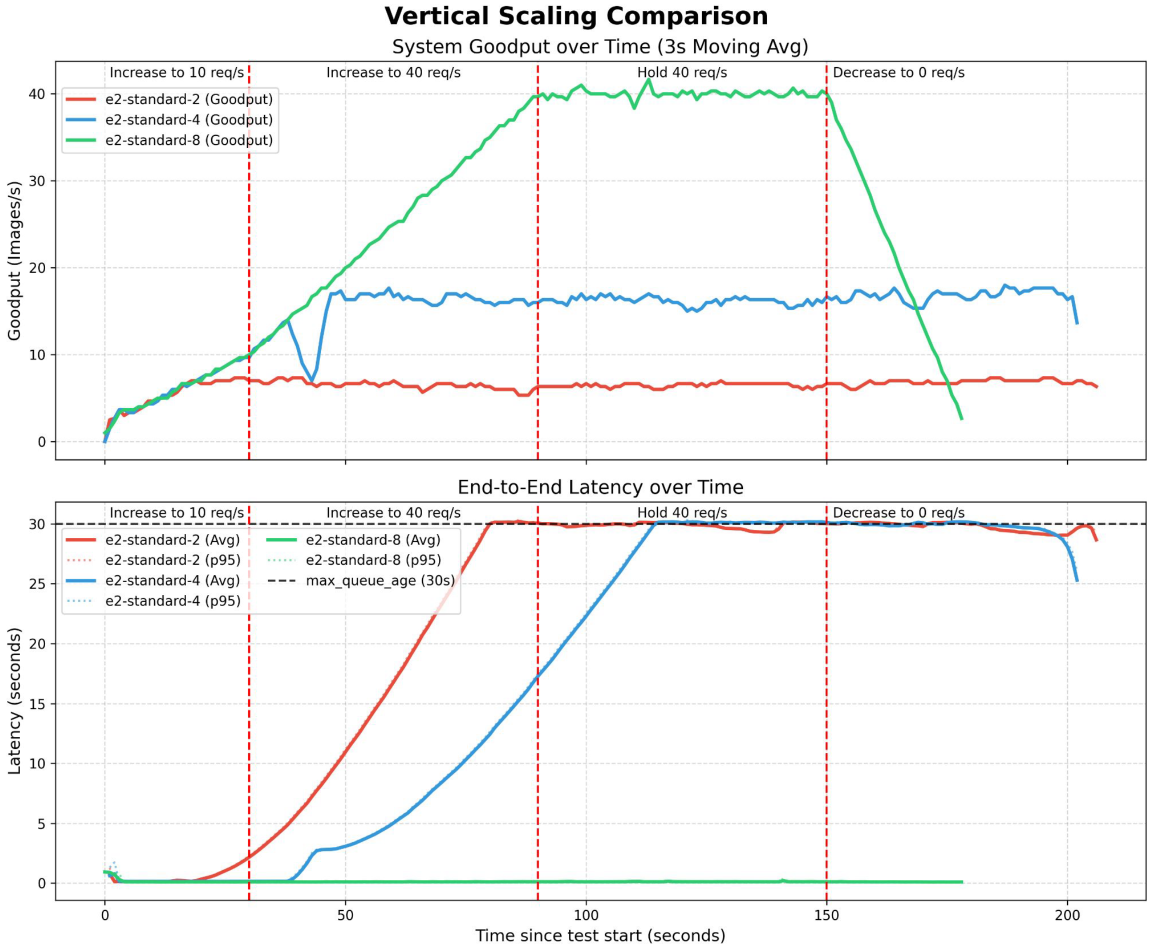
```
# from from app/api/main.py
if queue_length >= MAX_QUEUE_SIZE:
    return Response(content="Queue is full. Please retry later.", status_code=503)
```

Results

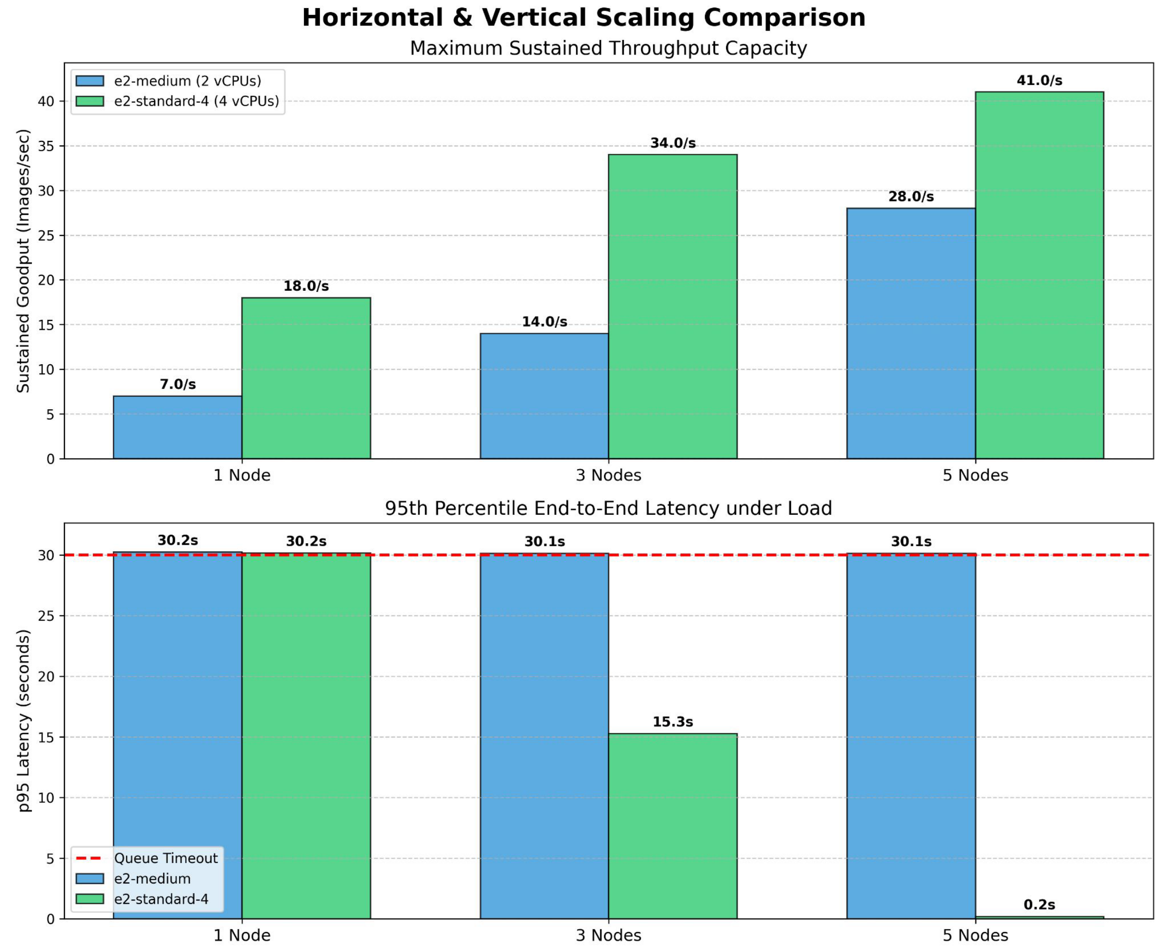


Horizontal-med Scaling Comparison
System Goodput over Time (3s Moving Avg)
Increase to 10 req/s
Increase to 40 req/s
Hold 40 req/s
Decrease to 0 req/s
1 Node (Goodput)
3 Nodes (Goodput)
5 Nodes (Goodput)
Goodput (Images/s)
0
5
10
15
20
25
30
End-to-End Latency over Time
1 Node (Avg)
1 Node (p95)
3 Nodes (Avg)
3 Nodes (p95)
5 Nodes (Avg)
5 Nodes (p95)
max_queue_age (30s)
Latency (seconds)
0
50
100
150
200
Time since test start (seconds)

Results


Vertical Scaling Comparison
System Goodput over Time (3s Moving Avg)
Increase to 10 req/s
Increase to 40 req/s
Hold 40 req/s
Decrease to 0 req/s
e2-standard-2 (Goodput)
e2-standard-4 (Goodput)
e2-standard-8 (Goodput)
Goodput (Images/s)
0
10
20
30
40
End-to-End Latency over Time
Increase to 10 req/s
Increase to 40 req/s
Hold 40 req/s
Decrease to 0 req/s
e2-standard-2 (Avg)
e2-standard-2 (p95)
e2-standard-4 (Avg)
e2-standard-4 (p95)
e2-standard-8 (Avg)
e2-standard-8 (p95)
max_queue_age (30s)
Latency (seconds)
0
5
10
15
20
25
30
0
50
100
150
200
Time since test start (seconds)

Results



Horizontal & Vertical Scaling Comparison
Maximum Sustained Throughput Capacity
e2-medium (2 vCPUs)
e2-standard-4 (4 vCPUs)
Sustained Goodput (Images/sec)
0
5
10
15
20
25
30
35
40
7.0/s
18.0/s
14.0/s
34.0/s
28.0/s
41.0/s
1 Node
3 Nodes
5 Nodes
95th Percentile End-to-End Latency under Load
p95 Latency (seconds)
0
5
10
15
20
25
30
30.2s
30.2s
30.1s
15.3s
30.1s
0.2s
Queue Timeout
e2-medium
e2-standard-4
1 Node
3 Nodes
5 Nodes

Limitations

- Single Point of Failure (Node 1)
- Dynamic Scaling
- Hyperparamters
  - MAX_API_CAPACITY
  - DB_MAX_CONN
  - THROUGHPUT
  - MAX_QUEUE_AGE

Thank you for listening :)

Questions?